IP addressing & Subnetting:-

Tasks:-

1. List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format. Ans.:-

IP Class	Address Range (First Octet)	Default Subnet Mask	Purpose	Real-World Example
Class A	1 – 126	255.0.0.0 (/8)	Very large networks	A large multinational company (e.g., a global enterprise with thousands of devices). Example IP: 10.1.1.1 (private Class A).
Class B	128 – 191	255.255.0.0 (/16)	Medium-sized networks	A university or large hospital network. Example IP: 172.16.5.10 (private Class B).
Class C	192 – 223	255.255.255.0 (/24)	Small networks	A home Wi-Fi router or small office network. Example IP: 192.168.1.100 (private Class C).

Class D	224 – 239	N/A	Multicast communication	IPTV streaming, live video conferencing, or online webinar broadcasts. Example multicast IP: 239.255.255.250**.**
Class E	240 – 255	N/A	Experimental and research use	Used for networking research and testing; generally not assigned to normal devices. Example IP: 240.0.0.1**.**

**2. Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses.

Hint: Show your working steps for each calculation.**

Ans.:-

Class C: 192.168.0.0 **IP**

Address: 192.168.10.45

subnet mask: 255.255.255.0

CIDR : 24

Network = 24 , **Host =** 8

No of Host :- 2^8 -2 = 256-2 =254

First Host Address : 192.168.10.1

Last Host Address: 192.168.10.254

Item	Value
IP Address	192.168.10.45
Subnet Mask	255.255.255.0 (/24)
Item	Value
Network Address	192.168.10.0
Broadcast Address	192.168.10.255
First Valid Host	192.168.10.1
Last Valid Host	192.168.10.254
Total Valid Hosts	254

3. Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

Ans.:-

Original Network: 10.0.0.0/24

Required Subnets: 4

1111111.11111111.11000000.00000000

2N

Network: $\mathbf{2^2 = 4}$

Subnet Mark =126+64=192

Subnet Mask: 255.255.255.192 (/26)

No of Host : $2^H - 2 = 2^6 - 2 = 64 - 2 = 62$

- Delivery Partners: 10.0.0.0/26

- Restaurants: 10.0.0.64/26
- Customers: 10.0.0.128/26
- Admins: 10.0.0.192/26

**4. You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet.

Constraint: Show your calculation steps and final answers.**

Ans.:-

Network Address: 172.16.0.0/16

Original Subnet Mask: 255.255.0.0

Required Subnets: 20

Use the formula: 2n≥Required Subnets

: $\mathbf{2^5 = 32}$

Original prefix = 16

Borrowed bits = 5

: 16+5=21

New CIDR Notation = 21

11111111.11111111.11111000.00000000

New Subnet Mask = 255.255.248.0

No of Network :-

= $\mathbf{2^5 = 32}$

No of Host :-

= $2^{11} - 2 = 2048 - 2 = \mathbf{2046}$

- **New Subnet Mask**: 255.255.248.0 (/21)
- **Number of Subnets**: 32
- **Hosts per Subnet**: 2046

5. A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

Ans.:-

The IP address 192.168.1.300 is invalid because each octet in an IPv4 address can only have a value from 0 to 255. Since 300 is greater than 255, the device cannot use this IP address and will not connect to the network.

If the subnet is 192.168.1.0/24 (subnet mask 255.255.255.0), a valid host IP address should be in the range 192.168.1.1 to 192.168.1.254. The address 192.168.1.0 is the network address, and 192.168.1.255 is the broadcast address, so neither can be assigned to a device.

For example, valid IP addresses include:

- 192.168.1.10
- 192.168.1.50
- 192.168.1.100
- 192.168.1.200

Changing the device's IP address to any valid host address within this range should allow it to connect to the network, provided there are no duplicate IP addresses or other network configuration issues.